MIN KHANT

AI ENGINEER | PYTHON BACKEND DEVELOPER
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SUMMARY

AI Engineer with 3+ years in machine learning and AI, including 1.5 years building and running production Large Language Model (LLM) systems. Designed and shipped a Retrieval-Augmented Generation (RAG) customer assistant for a Singapore precious-metals trader, evaluated four model options before selecting one for production, and re-architected it as a tool-calling ReAct agent to fix hallucinations and tool-calling failures. Backed by strong production backend skills in Python, FastAPI, PostgreSQL, Docker, AWS, and Azure, including a 97% API latency reduction, plus earlier Natural Language Processing (NLP) and computer vision work for Japanese public-sector clients.

SKILLS

RAG | AI Agents | ReAct | Tool Calling | Function Calling | Prompt Engineering | LLM Evaluation | Fine-tuning | Quantization | LangChain | LangGraph | LangSmith | HuggingFace | Transformers | OpenAI API | Ollama | Embeddings | Vector Databases | ChromaDB | LlamaIndex | Chunking Strategies | Context Management | Semantic Search | Advance Retriving Techniques | PyTorch | TensorFlow | Scikit-Learn | Pandas | Numpy | CNN | NLP | Computer Vision | Python | FastAPI | RESTful APIs | Celery | Webhooks | SQL | JavaScript | React.js | Node.js | PostgreSQL | MongoDB | Redis | AWS | Azure | Docker | Version Control | CI/CD | GitHub Actions | Git | Unit Testing | Load Testing.

WORK EXPERIENCE

Backend Developer (Freelance) | Canopus Can-Trading Co., Ltd.
08/2025 - Present | Singapore (Remote)

- Built and maintained a Python/FastAPI backend service for 5-minute vessel data ingestion, business logic, and RESTful APIs supporting **over 400 vessels** without crashing the server.
- Architected the system into controller, service, and repository layers, making new features faster to build and test.
- Engineered validation and transformation pipelines for third-party data sources, improving data consistency across the platform.
- Optimized PostgreSQL queries and schema usage, and added a cache layer after role-based access control (RBAC) validation, reducing data retrieval latency by **97% (~5s to ~200ms)**.
- Deployed services on AWS with Docker and Nginx, hardening them with security middleware, rate limiting, and restricted network rules.

AI Engineer (Full-time) | GoldSilver Central Co., Ltd.
05/2024 - 11/2025 | Singapore (Remote)

- Designed and shipped a **RAG pipeline** powering an **LLM** WhatsApp customer assistant that grounds answers in company documents, serving over **100 users**.
- Evaluated four candidate approaches (**fine-tuned** and **quantized open-source models**, **Llama 2 Instruct**, and the **OpenAI Responses API**) on accuracy, latency, maintainability, and hallucination rate; selected the OpenAI Responses API for production.
- Diagnosed hallucination and tool-calling failures in production and re-architected the assistant as a **ReAct agent**.
- Built FastAPI services and webhook-driven automation on Azure that cut the team's manual workload by **90%** by automating routine price and order-status inquiries.
- Translated business requirements from non-technical stakeholders into modular, extensible AI workflows, working in a 2-developer team.

AI Engineer (Full-time) | Doctoral Institute for Evidence-Based Policy, Inc.（政策基礎研究所
06/2023 - 05/2024 | Yangon, Myanmar (Hybrid)

- Built a fingerprint classification system, benchmarking rule-based minutiae matching, **CNN classifiers**, and Siamese networks; the best model reached **95% accuracy**.
- Developed end-to-end **BERT workflows** (preprocessing, fine-tuning, evaluation) for Japanese-language sentiment analysis, reaching **~80% accuracy** on binary class labels.

- Resolved Unicode normalization issues (half-width vs. full-width Japanese characters) across large datasets, improving training-data consistency.
- Created reusable preprocessing pipelines and visualization tools, speeding up experimentation and model debugging.
- Collaborated with engineering teams in Japan to integrate trained models into production web applications.

PROJECTS

Alferd | RAG Document Assistant
github.com/MinKhantYCC/alferd-backend | github.com/MinKhantYCC/alferd-frontend
- Built a full-stack RAG application for PDF question answering with FastAPI, LangChain, Ollama, and React.js.
- Offloaded document processing to Celery background tasks, reducing response latency by **75%**;
- Streamed model response for better user experience and tracked LLM usage and performance with LangSmith.
- Verified reliability with PyTest unit tests and Locust load tests.

Loan Repayment Prediction — Machine Learning Classifiers
kaggle.com/code/linyaungzin/loan-prediction-rf/notebook
- Built and analyzed Scikit-learn classification models to predict loan repayment using structured financial data.
- Performed preprocessing, feature engineering, model training, cross-validation, and performance analysis with Pandas, Scikit-learn, and Matplotlib.
- Attained approximately **77% test accuracy** while developing practical experience in financial data modeling and ML-based decision support.

Burmese - English Translation Model
medium.com/@linyaungzin99/fine-tuning-metas-nllb-200-1-3b-e83235a238b4
- Fine-tuned Meta's NLLB-200-1.3B model on 12,021 Burmese-English sentence pairs using 4-bit quantization and LoRA, improving SacreBLEU from 11.94 to 26.94 (+125.6%) over the baseline.
- Optimized GPU memory efficiency using NF4 quantization, rank-8 LoRA adapters, and dynamic padding, enabling fine-tuning of a 1.3B-parameter model on a single NVIDIA T4 GPU (15 GB VRAM).
- Developed and evaluated an end-to-end machine translation pipeline using Hugging Face Transformers, PEFT, and PyTorch, completing 2 training epochs in approximately 62 minutes and validating performance on 1,203 held-out test samples.

EDUCATION

B.Sc. Computer Science | University of the People (online) | 08/2024 – 06/2027 (expected)
B.E. Electronics and Communications | University of Technology (Yatanarpon Cyber City) | 12/2015 – 03/2020

CERTIFICATIONS

Machine Learning Specialization | Coursera | Jan 2023
Deep Learning Specialization | DeepLearning.AI | Jul 2023
Associate Data Scientist | DataCamp | Jun 2024
Python Django - The Practical Guide | Udemy | Jan 2026
Node.js for Beginners | Packt | May 2026
Developing Front-End Apps with React | IBM | Sep 2026

LANGUAGES

Burmese (native) | **English** (C1, advanced) | **Japanese** (JLPT N4)